MIL-C-24491B(SH)
21 April 1981
SUPERSEDING
MIL-C-24491A(SHIPS)
14 June 1974
(See 6.5)

MILITARY SPECIFICATION

CONTAINER, FIBERGLASS REINFORCED PLASTIC, FOR INFLATABLE LIFEBOATS, MARK 5, MOD 2, AND MARK 6

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers stowage container intended for 15-person Mark 5, Mod 2 and 25-person Mark 6, inflatable lifeboats.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

QQ-B-613 - Brass, Leaded and Nonleaded: Flat Products (Plate, Bar, Sheet and Strip).
QQ-B-626 - Brass, Leaded and Nonleaded: Rod, Shapes, Forgings, and Flat Products with Finished Edges (Bar and Strip).
PPP-B-636 - Boxes, Shipping, Fiberboard.
PPP-B-640 - Boxes, Fiberboard, Corrugated, Triple-Wall.

MILITARY

MIL-P-116 - Preservation-Packaging, Methods of.
MIL-R-6130 - Rubber, Cellular, Chemically Blown.
MIL-P-17549 - Plastic Laminates, Fibrous Glass Reinforced, Marine Structural.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Sea Systems Command, SEA 3112, Department of the Navy, Washington, DC 20362 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 1940

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MILITARY - Continued

MIL-R-21607 - Resins, Polyester, Low Pressure Laminating, Fire Retardant.
MIL-C-43307 - Cord, Nylon, Solid Braid, General Purpose.
MIL-I-45208 - Inspection System Requirements.

STANDARDS

FEDERAL

FED-STD-595 - Colors.

MILITARY

MIL-STD-129 - Marking For Shipment and Storage.

DRAWING

NAVAL SEA SYSTEMS COMMAND (NAVSEA)

NAVSHIPS 803-4382176 - Container, Fiberglass Reinforced Plastic, for Inflatable Lifeboats.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of the specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitations for bids or request for proposal shall apply.

NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION, INC. AGENT

National Motor Freight Classification

(Application for copies should be addressed to the National Motor Freight Traffic Association, Inc., ATA TRAFFIC Dept., 1616 "P" Street, N.W., Washington, DC 20036.)

UNIFORM CLASSIFICATION COMMITTEE AGENT

Uniform Freight Classification Ratings, Rules, and Regulations

(Application for copies should be addressed to the Uniform Classification Committee Agent, Tariff Publication Officer, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 First article sample. Prior to beginning production, one sample container shall be examined and tested as specified in 4.3 (see 6.3).

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3.2 Materials. Materials used in the construction of the container shall be in accordance with table I.

TABLE I. Material.

Item	Material	Applicable document
Container body (see 3.3.2)	Fibrous glass reinforced plastic	MIL-R-21607, grade 1, fire resistant For layup schedule see Drawing 803-4382176.
Gasket (see 3.3.3)	Cellular rubber, oil and flame resistant	MIL-R-6130, type I, grade A, condition medium
Banding straps (see 3.3.4)	UNS 26000 brass strip, annealed temper	QQ-B-613 QQ-B-626
Painter (see 3.3.5)	Braided nylon cord	MIL-C-43307
Container grommet (see 3.3.6)	Synthetic rubber	Drawing 803-4382176
Joint seal (see 3.3.7)	Synthetic rubber	

3.2.1 Recovered materials. Unless otherwise specified herein, all equipment, material, and articles incorporated in the products covered by this specification shall be new and shall be fabricated using materials produced from recovered materials to the maximum extent practicable without jeopardizing the intended use. The term "recovered materials" means materials which have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of used or rebuilt products is allowed under this specification unless otherwise specifically specified.

3.3 Design and construction. Dimensions, shape, and general arrangement of the container shall conform to Drawing 803-4382176.

3.3.1 Weight. The total weight of the container assembly including all fittings shall not exceed 70 pounds.

3.3.2 Container body. Configuration of the container body shall be as shown on Drawing 803-4382176 with mating top and bottom halves. The thickness of the plastic laminate shall be as required to meet the examination and tests specified herein.

3.3.3 Gasket. A 1/4-inch thick gasket configured to the mating flange of the bottom container half and bonded in place with a waterproof adhesive shall be provided on the bottom half of the container.

3.3.4 Banding straps. Banding straps shall be 3/4-inch wide by approximately 0.020-inch thick and shall have one 3/32-inch diameter hole in each band such that the breaking strength of each band will withstand the test of 4.6.2 and will break from the force exerted by inflation of the lifeboat without causing damage to the container.

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3.3.5 Painter. A 5/32-inch diameter, 100-foot length, nylon painter with a breaking strength of not less than 540 pounds coiled in a cylindrical cardboard tube for free running shall be provided. One end of the painter shall remain in the container for attachment to the release mechanism and dee ring of the stowed boat. The other end shall be free and extend approximately 1 foot through the painter cork for attachment to ship structure. The ends of the painter shall be seized. The outside free end of the painter shall be temporarily taped to case to prevent snagging of the painter during handling.

3.3.6 Grommet. A two-piece container grommet conforming to details 5A and 5B of Drawing 803-4382176 shall be provided. The grommet shall be snugly installed into a U shaped cutout, open at the joint between container halves in the bottom half of the container.

3.3.7 Joint seal. After the installation of banding, the joint between the top and bottom halves of the container shall be covered with a 6-inch wide rubber band seal, B.F. Goodrich part number 1RA1056, or equal.

3.3.8 When specified (see 6.2.1), when containers are shipped empty, the banding straps, painter and painter stowage tube, and sealing band are to be packaged and shipped separate from the containers as specified in 5.2.2.

3.4 Finish. The exterior surfaces of the container shall have a pigmented gel coat in accordance with MIL-P-17549, 5 mil to 25 mil thickness. Gel coat and laminating resins shall be pigmented with color conforming to 26307 of FED-STD-595. Interior surfaces shall be smooth with no exposed laminate fibers and free of sharp or jagged edges.

3.5 Marking. Except as otherwise specified in 3.5.1.3 and 3.5.1.4 (see 6.2.1), the markings shall be legible and durable black Gothic letters and numerals which shall be thoroughly dry prior to packaging. The ink used shall have an acidity of 5.0 to 8.5 and shall contain no ingredient which is deleterious to the material being marked.

3.5.1 Container marking. The container shall contain the markings specified herein and located as shown on Drawing 803-4382176.

3.5.1.1 Top half. The container side shall contain the following information in 1/4-inch black letters and numerals:

"INFLATABLE LIFEBOAT MK ____* , MOD ____* , MAXIMUM CAPACITY ____* PERSONS
CONTAINER, INFLATABLE LIFEBOAT
NSN #
BUILT FOR U.S. NAVY
CONTRACTOR'S NAME, CITY _____#_____
MONTH, YEAR CONTRACT NO. _____#_____ "

\# To be completed by container manufacturer.
* To be completed by boat manufacturer or packing activity.

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3.5.1.2 Upper and lower half. The upper half of the container shall be labeled "Top" and the lower half "Bottom." The letter size shall be 1-inch.

3.5.1.3 End. At the end, where the container grommet is located, in 1/4-inch high red letters above the grommet, the top half of the container shall be marked:

"HANDS OFF
EXCEPT IN EMERGENCY
TO INFLATE:
LAUNCH CONTAINER, PULL PAINTER
TO DEPLOY EXCESS LINE, TUG
PAINTER HARD TO ACTIVATE.
DO NOT
INFLATE ON HARD SURFACE."

3.5.1.4 Directly below the identification markings (see 3.5.1.1) and approximately 3-1/2 inches above the joint between container halves, in 1/2-inch red gothic letters, the container shall be marked in red as follows:

"DO NOT ROLL THIS CONTAINER."

3.6 Workmanship. Workmanship shall be such as to assure that the product is free of defects that would damage the inflatable lifeboat during stowage and inflation.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Inspection system. The contractor shall provide and maintain an inspection system acceptable to the Government for supplies and services covered by this specification. The inspection system shall be in accordance with MIL-I-45208.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

(a) First article inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).

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4.3 First article inspection. First article inspection shall consist of the examination specified in 4.5 and the tests specified in 4.6.1 and 4.6.2. One sample container along with test panel specified in MIL-P-17549 shall be forwarded to the Officer-in-Charge, David W. Taylor Naval Ship Research and Development Center, Annapolis Laboratory, Annapolis, MD 21402. The sample container shall be representative of the construction, workmanship, components and materials to be used during production.

4.3.1 First article test reports. The contractor shall prepare a first article test report in accordance with the data ordering document included in the contract or order (see 6.2.2).

4.4 Quality conformance inspection. Quality conformance inspection shall consist of the following:

(a) Visual and dimensional examination (see 4.5).
(b) Marking (see 3.5 and 5.3).

4.5 Examination. Each container shall be thoroughly checked and visually and dimensionally examined to determine conformance to the requirements of this specification not involving tests. Table II shall be used to classify the critical defects. No container will be accepted until all discernible defects are corrected.

TABLE II. Classification of defects.

Categories	Defects
Critical	
1	Weight of case in excess of 70 pounds.
2	Dimensions not in accordance with Drawing 803-4382176.
3	Two halves of container do not seal tightly.
4	Inner surface not smooth; has rough or sharp projections.
5	Gasket missing or improperly installed.
6	Closed container not watertight.
7	Gel coat pitted, crazed, or cracked.
8	Handling recesses too small, improperly located or omitted.
9	Nylon painter missing or improperly coiled.
10	Incorrect marking or instructions.
11	Drain holes omitted.
12	Color not as required.
13	Evidence of unauthorized material or repairs.
14	Fiberglass layup not in accordance with Drawing 803-4382176.
Major	None defined.
Minor	None defined.

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4.6 Tests.

4.6.1 Watertightness. The assembled container placed in any position shall be subjected to a water stream directed around the entire sealing area from a hose equipped with a 1/2-inch diameter (minimum) nozzle under not less than 25 pounds per square inch (lb/in^2) line pressure at a distance of not more than 10 feet from the case for a period of 10 minutes. There shall be no evidence of water entering the container.

4.6.2 Drop test. The assembled container with a Mark 6 lifeboat and all equipment and survival gear within shall be dropped into water from a height of not less than 65 feet. The container and lifeboat shall remain afloat for not less than 30 minutes after the drop and the lifeboat shall inflate as designed. The lifeboat used in this test shall be packed with its center-of-gravity located so that the container rotates to the upright position (plus or minus 10 degrees). Damage caused by container striking the water with the mating joint vertical shall not be cause for rejection, but the test must be repeated with the lifeboat properly packed.

4.7 Inspection of preparation for delivery. Packaging, packing, and marking shall be inspected for compliance with section 5 of this specification.

5. PACKAGING

(The preparation for delivery requirements specified herein apply only for direct Government acquisitions. For the extent of applicability of the preparation for delivery requirements of referenced documents listed in section 2, see 6.4.)

5.1 General requirements. Unless otherwise specified (see 6.2.1), containers shall be knocked-down and nested for shipment and storage.

5.2 Packaging and packing. Packaging and packing shall be level A or C, as specified (see 6.2.1).

5.2.1 Level A. Each container and its accessories (see 3.3.4 through 3.3.7) shall be packaged as specified in method III of MIL-P-116 and packed as follows:

(a) The container halves shall be nested one inside the other by placing the bottom inside the top.
(b) Prior to nesting the container halves, the accessories shall be placed inside the container top and secured; alternatively, the accessories may be wrapped in a noncorrosive barrier material, secured with tape and placed in the top half of the container. Pressure sensitive filament reinforced tape shall be used to secure the container halves together.
(c) A strip of cushioning material shall be placed between the mating areas of the container halves prior to packing the container.
(d) Each container nested as specified, shall be packed in a fiberboard box conforming to PPP-B-640, class 2 or PPP-B-636, class weather-resistant, grade V.

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(e) The fiberboard box shall provide support to all panels as required in the appendix to PPP-B-636 to provide a type Z average load as specified therein. Box reinforcement and closure shall be in accordance with the appendix to the box specification except for PPP-B-636 boxes, method V of the appendix to the box specification shall apply.

5.2.2 Level C. Each container and its accessories (see 3.3.4 through 3.3.7), shall be packaged and packed to afford protection against loss of contents, deterioration and physical damage during shipment from the supply source to the first receiving activity for immediate use. Containers used or method of packing shall be acceptable to the common carrier which will insure safe delivery at destination in a satisfactory condition at the lowest applicable rate and shall comply with Uniform Freight or National Motor Freight Classification Rules and Regulations or other carrier rules as applicable to the mode of transportation.

5.3 Marking. In addition to any special markings required (see 6.2.1), the packing containers shall be marked in accordance with MIL-STD-129 and shall include the date of manufacture (month and year).

6. NOTES

6.1 Intended use. The containers covered by this specification are intended to be used for shipboard stowage of Mark 5, Mod 2 and Mark 6 inflatable lifeboats.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

(a) Title, number, and date of this specification.
(b) When accessories are to be packaged separate from the container (see 3.3.8).
(c) Marking required (see 3.5 and 5.3).
(d) When containers are not to be nested (see 5.1).
(e) Level of packaging and packing required (see 5.2).

6.2.2 Data requirements. When this specification is used in a contract which incorporates a DD Form 1423 and invokes the provisions of 7-104.9(n) of the Defense Acquisition Regulation (DAR), the data requirements identified below will be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved Contract Data Requirements List (DD Form 1423) incorporated into the contract. When the provisions of DAR-7-104.9(n) are not invoked, the data specified below will be delivered by the contractor in accordance with the contract requirements. Deliverable data required by this specification is cited in the following paragraphs:

Paragraph	Data requirement	Applicable DID	Option
4.3.1	First article inspection report	DI-T-4902	----

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(Copies of data item descriptions required by the contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

6.2.2.1 The data requirements of 6.2.2 and any task in section 3, 4, or 5 of the specification required to be performed to meet a data requirement may be waived by the contracting/acquisition activity upon certification by the offeror that identical data were submitted by the offeror and accepted by the Government under a previous contract for identical item acquired to this specification. This does not apply to specific data which may be required for each contract, regardless of whether an identical item has been supplied previously (for example, test reports).

6.3 First article inspection. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection as to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

6.4 Sub-contracted material and parts. The preparation for delivery requirements of referenced documents listed in section 2 do not apply when material and parts are acquired by the contractor for incorporation into the equipment and lose their separate identity when the equipment is shipped.

6.5 Changes from previous issue. Asterisks (*) are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

* U.S. GOVERNMENT PRINTING OFFICE: 1991—703-023/22273

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MIL-C-24491B (SH)

NAME OF ORGANIZATION AND ADDRESS

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MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

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